

NORM

Let $F = \mathbb{C}$ or $\mathbb{R}$. A vector space $\mathcal{V}$ over F equipped with a norm is called a normed vector space. A norm is a function

$$\| \cdot \| : \mathcal{V} \rightarrow \mathbb{R}^+$$

such that-

- (i) $\| \alpha \cdot v \| = |\alpha| \| v \|$, $\forall \alpha \in F, v \in \mathcal{V}$
- (ii) $\| v \| \geq 0$, $\forall v \in \mathcal{V}$, and $\| v \| = 0$ if & only if $v = 0$
- (iii) $\| u + v \| \leq \| u \| + \| v \|$, $\forall u, v \in \mathcal{V}$
(Triangle Inequality)

A vector space $\mathcal{V}$ equipped with a norm is called a normed vector space.

Normed Vector Spaces

A vector space $\mathcal{V}$ equipped with a norm, is called a normed vector space.

Examples

① $\mathcal{V} = \mathbb{R}^n$ or $\mathbb{C}^n$: Define $\|\cdot\|_p: \mathcal{V} \rightarrow \mathbb{R}^+$ as

$$\|v\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{\frac{1}{p}}, \quad v \in \mathcal{V}$$

It can be shown that $\|\cdot\|_p$ is a norm!
(non-trivial to verify triangle inequality except when $p=2$)
for $1 \leq p \leq \infty$

Examples:

(i) l_1 norm: $\|v\|_1 = \sum_{i=1}^n |v_i|$

(ii) l_2 norm: $\|v\|_2 = \left(\sum_{i=1}^n |v_i|^2 \right)^{\frac{1}{2}}$

[can be related to the inner product]

$$\|v\|_2 = \sqrt{\sum_{i=1}^n |v_i|^2} = \sqrt{\sum_{i=1}^n \bar{v}_i v_i} = \sqrt{\langle v, v \rangle}$$

inner product in $\mathbb{C}^n$

(iii) l_∞ norm:

$$\|v\|_\infty = \max_{i=1,2,\dots,n} \{ |v_i| \}$$

② $\mathcal{V} = l_p$:- $1 \leq p \leq \infty$

$$l_p = \left\{ \{x_n\}_{n \geq 1} \mid \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}$$

Define

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{\frac{1}{p}}$$

Can be shown that this is a norm.

③ $\mathcal{V} = L_p([a, b])$:- $1 \leq p \leq \infty$

$$L_p([a, b]) = \left\{ f : [a, b] \rightarrow \mathbb{R}, \text{ s.t. } \int_a^b |f(x)|^p dx < \infty \right\}$$

Define

$$\|f\|_p = \left(\int_a^b |f(x)|^p dx \right)^{\frac{1}{p}}$$

Can be shown that this is a norm!

Norm Induced By Inner Product.

Let V be equipped with inner product $\langle \cdot, \cdot \rangle$.

Define a function

$$\| \cdot \|_{ip} : V \rightarrow \mathbb{R}^+ \text{ as}$$

$$\|v\|_{ip} = \sqrt{\langle v, v \rangle}, \quad v \in V$$

Can be shown that:

$\| \cdot \|_{ip}$ is a norm.

[Using Cauchy-Schwarz inequality]

Examples:

also called
Euclidean Norm.

① $\mathbb{C}^n$: $\langle u, v \rangle = v^H u$.

$\Rightarrow \|u\|_{ip} = \sqrt{u^H u} = \left(\sum_{i=1}^n |u_i|^2 \right)^{\frac{1}{2}}$.

$\downarrow$
 l_2 -norm.

② l_2 = $\left\{ \{x_n\}_{n \geq 1}, \sum_{i=1}^{\infty} |x_i|^2 < \infty \right\}$

Can be shown that following is an inner product:

$$\langle u, v \rangle = \sum_{i=1}^{\infty} u_i \overline{v_i}$$

$$\|u\|_{ip} = \left(\sum_{i=1}^{\infty} |u_i|^2 \right)^{\frac{1}{2}}$$

③ $L_2([a, b])$: The following can be shown to be an inner product

$$\langle f, g \rangle = \int_a^b \overline{g(x)} f(x) dx$$

It induces the following

$$\|f\|_{L_2} = \left(\int_a^b |f(x)|^2 dx \right)^{\frac{1}{2}}.$$

ORTHOGONAL PROJECTION

[For this topic, we will assume to operate in a finite dimensional vector space over $\mathbb{F} = \mathbb{C}$ or $\mathbb{R}$. Generalizations of the results to ^{suitable} infinite dimensional vector spaces will be mentioned whenever relevant, but we will not prove them.]

Let V be a vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, endowed with an inner product $\langle \cdot, \cdot \rangle$ and let $\| \cdot \|_{\text{ip}}$ be the norm induced by this inner product. V is finite dimensional!

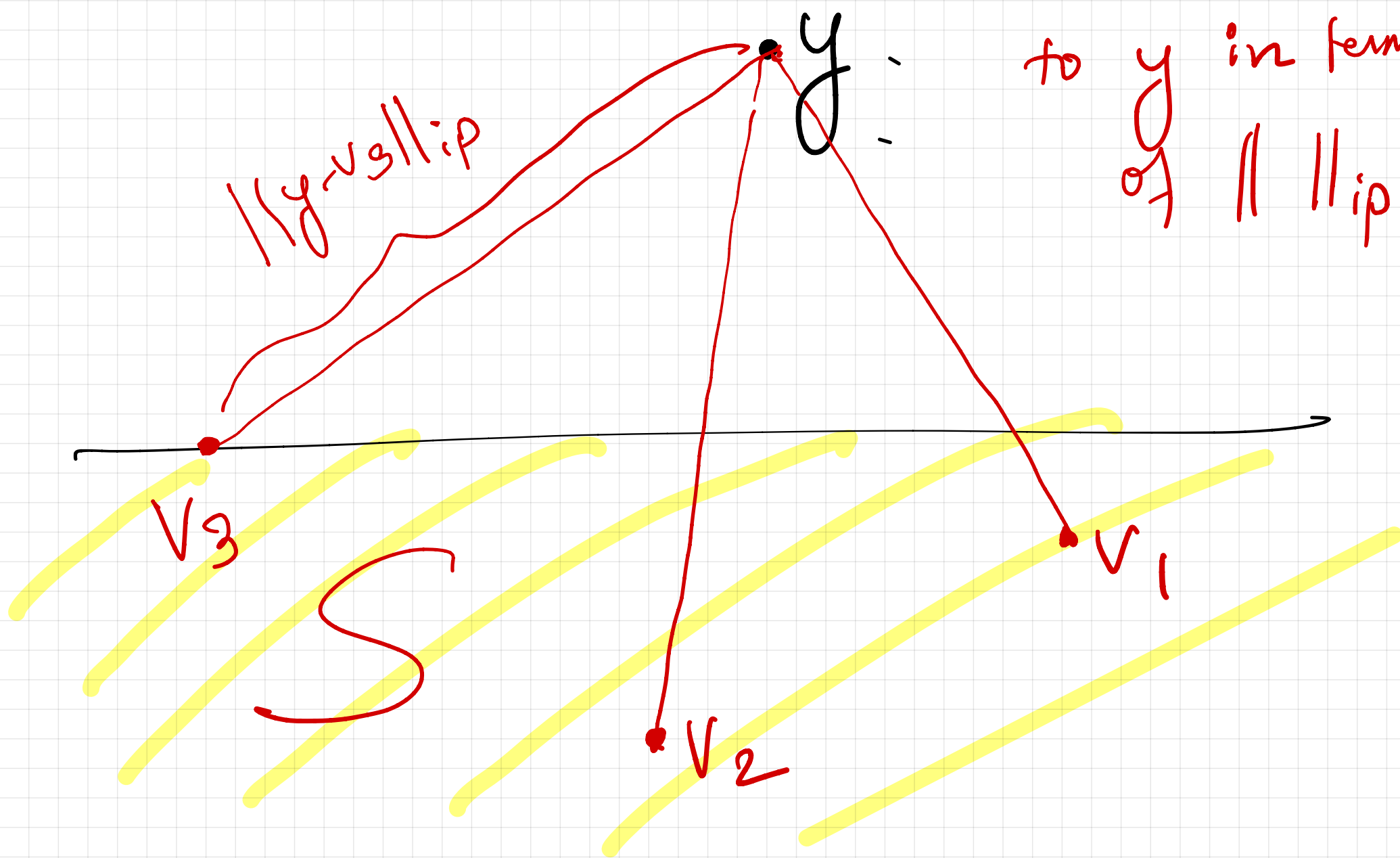
Let $y \in V$, and $S \subseteq V$ be a subspace

Consider the Optimization Problem:

$$\begin{aligned} \min_v & \|y - v\|_{ip} \\ \text{s.t. } & v \in S. \end{aligned} \quad \text{--- (P)}$$

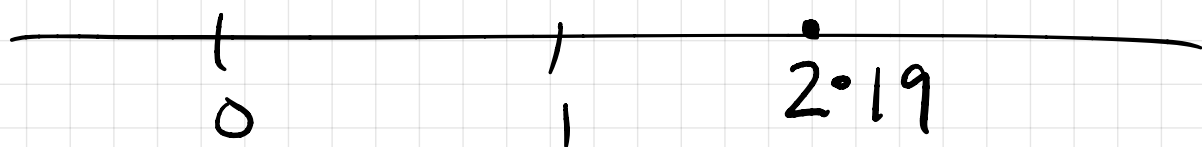
What is it trying to find?

A point in S which is "closest" to y in terms of $\|\cdot\|_{ip}$.



Can an optimization problem always be solved?

Does a minimizer always exist? *Not always*

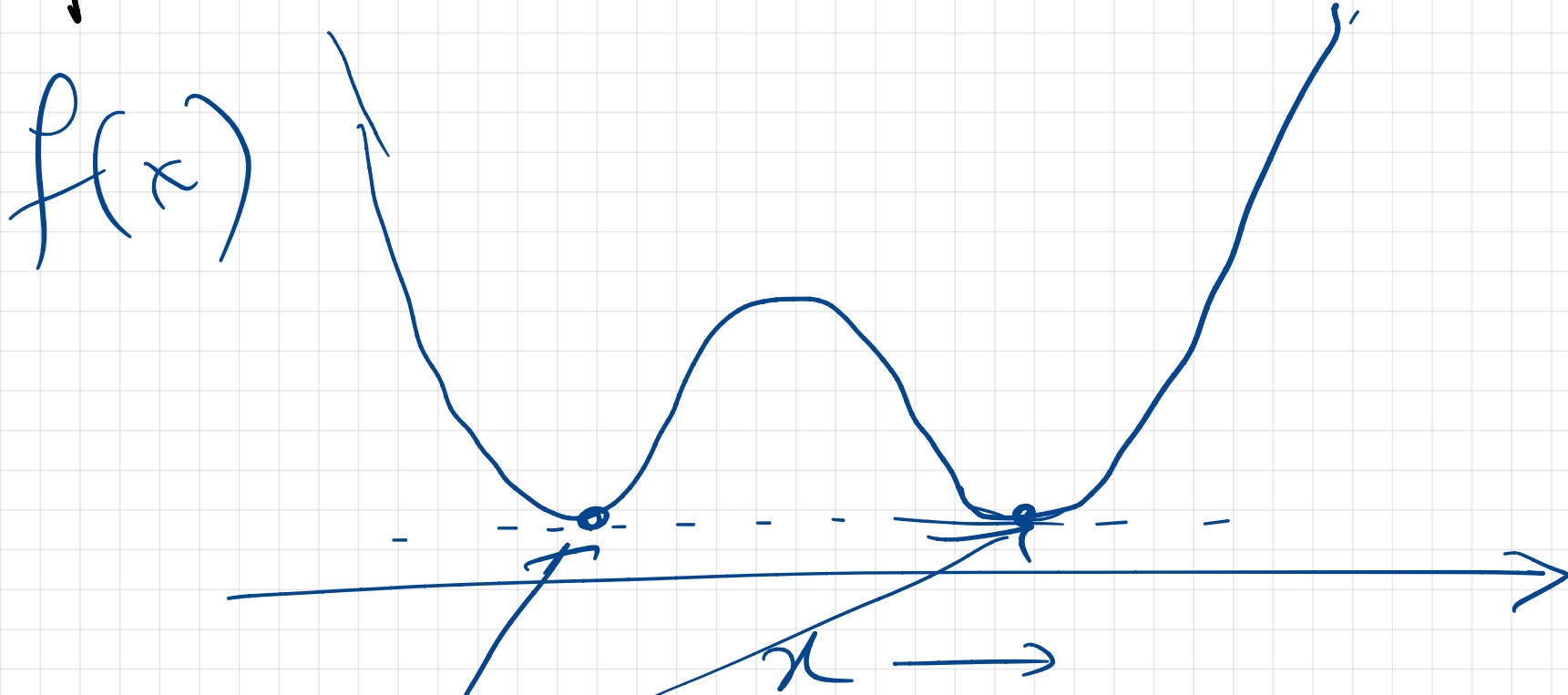


$$\min_x |2.19 - x| \quad \left. \begin{array}{l} 0 < x < 1 \end{array} \right\} \text{No minimizer}$$

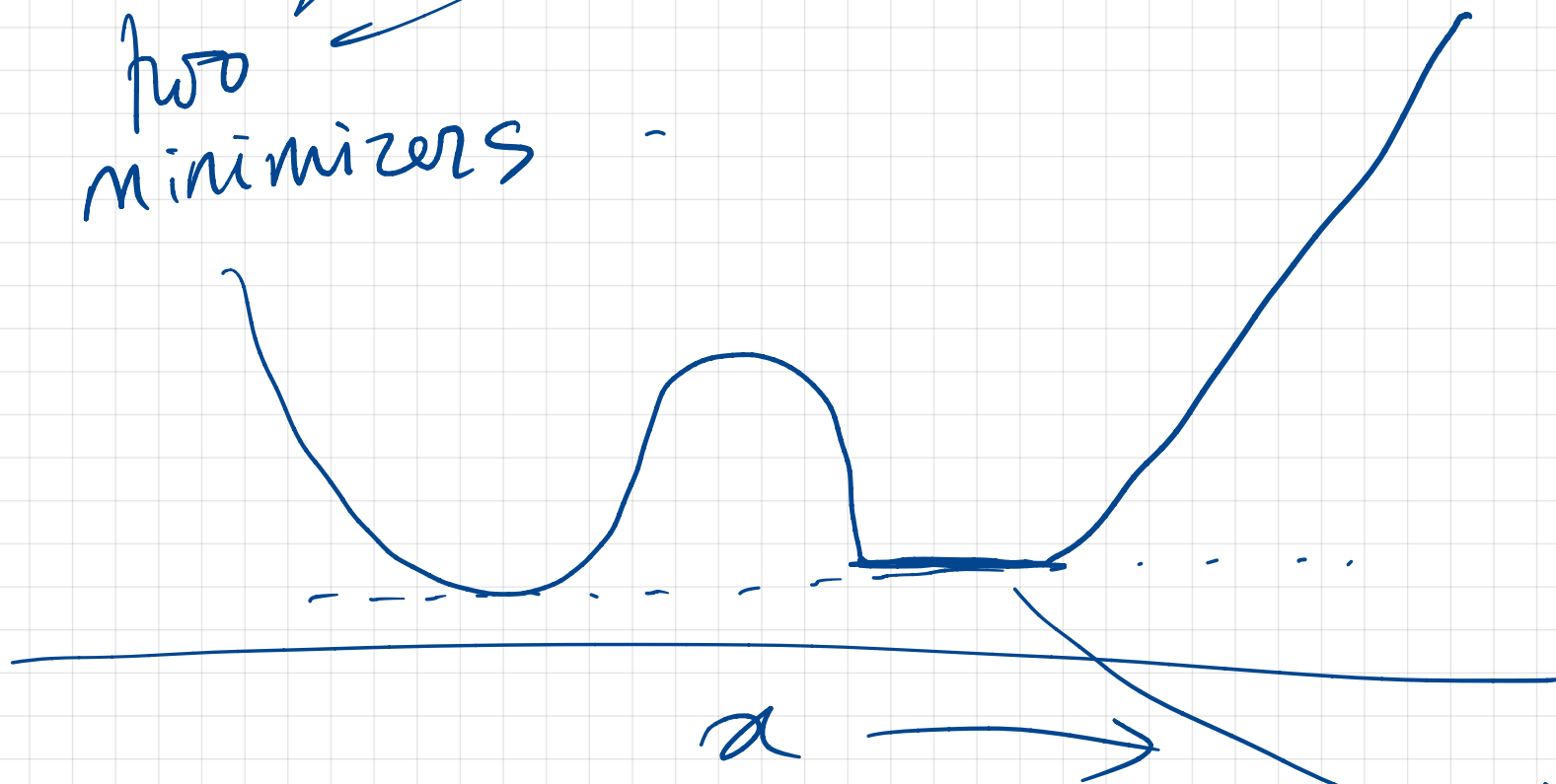
$$\min_x |2.19 - x| \quad \left. \begin{array}{l} 0 < x \leq 1 \end{array} \right\} \text{Minimizer exists} \\ x^* = 1.$$

We will show that the Problem (P)
has a minimizer (minimizer exists)

Suppose there is a solution to an optimization problem, is it unique?



two minimizers



infinite minimizers

We will show that (P) has a "unique" minimizer.

We will solve (P) without Calculus!

SUMMARY (WILL PROVE)

We will show that the minimizer to (P)

(i) exists

(ii) is unique.

(iii) Satisfies the orthogonality principle.

(will define soon) - This will lead to a computational algorithm to solve (P)

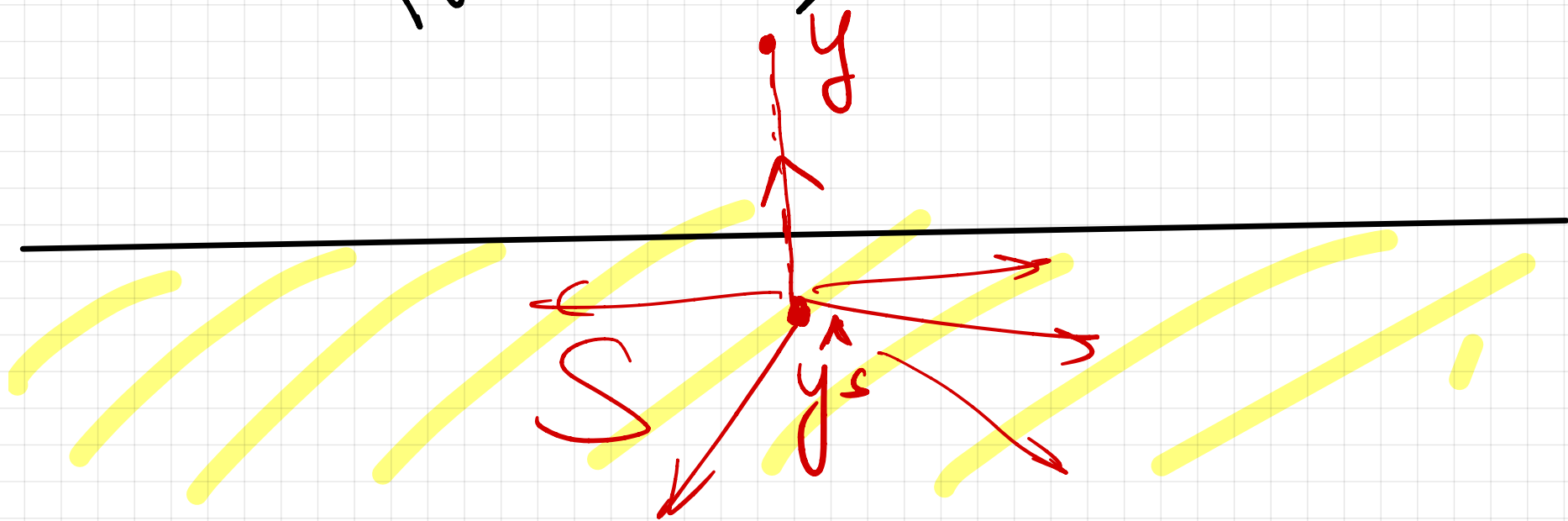
We will establish these facts by first studying "Orthogonal Projection"

Then we will show that Orthogonal projection is the minimizer of (P)

ORTHOGONAL PROJECTION

Given a point $y \in V$ and a subspace $S \subseteq V$,
a vector $\hat{y}_S \in S$ is called an orthogonal
projection of y onto S if

$$\langle y - \hat{y}_S, v \rangle = 0, \quad \forall v \in S$$



The existence and uniqueness of an orthogonal
projection can be established for an infinite
dimensional vector space V , if V is a Hilbert
Space (a complete inner product space) and S
is a closed subspace of V .

We will establish existence & uniqueness for
finite dimensional V .

UNIQUENESS OF ORTHOGONAL PROJECTION

Suppose there are two orthogonal projections of y onto S . Denote them by $\hat{y}_S, \hat{\hat{y}}_S$

We must have

$$(i) \quad \hat{y}_S \in S, \quad \langle y - \hat{y}_S, v \rangle = 0, \quad \forall v \in S$$

$$(ii) \quad \hat{\hat{y}}_S \in S, \quad \langle y - \hat{\hat{y}}_S, v \rangle = 0, \quad \forall v \in S$$

Consider

$$\begin{aligned} \|\hat{y}_S - \hat{\hat{y}}_S\|_p^2 &= \langle \hat{y}_S - \hat{\hat{y}}_S, \hat{y}_S - \hat{\hat{y}}_S \rangle \\ &= \langle \hat{y}_S - y + y - \hat{\hat{y}}_S, \hat{y}_S - \hat{\hat{y}}_S \rangle \\ &= \langle \hat{y}_S - y, \hat{y}_S - \hat{\hat{y}}_S \rangle + \langle y - \hat{\hat{y}}_S, \hat{y}_S - \hat{\hat{y}}_S \rangle \end{aligned}$$

Notice, $\hat{y}_S - \hat{\hat{y}}_S \in S$ ($\because \hat{y}_S, \hat{\hat{y}}_S \in S$ & S is subspace)

And we know

$$\langle \hat{y}_S - y, v \rangle = 0, \quad \langle y - \hat{\hat{y}}_S, v \rangle = 0 \\ \forall v \in S$$

$$\Rightarrow \langle \hat{y}_S - y, \hat{y}_S - \hat{\hat{y}}_S \rangle = 0$$

$$\& \langle y - \hat{\hat{y}}_S, \hat{y}_S - \hat{\hat{y}}_S \rangle = 0$$

$$\Rightarrow \|\hat{y}_S - \hat{\hat{y}}_S\|_{ip}^2 = 0$$

$$\Rightarrow \hat{y}_S = \hat{\hat{y}}_S.$$

Orthogonal projection of y onto S is

unique!

FINDING THE ORTHOGONAL PROJECTION

We will use the fact that V is finite dimensional. Therefore S is finite dimensional.

$$\text{Let } \dim(S) = k$$

Let $\{u_1, u_2, \dots, u_k\}$ be a basis for S .

Note that $\hat{y}_S$ is an orthogonal projection of y onto S iff $\hat{y}_S \in S$ and

$$\langle y - \hat{y}_S, v \rangle = 0, \quad \forall v \in S$$

$$\Leftrightarrow \langle y - \hat{y}_S, u_i \rangle = 0, \quad i = 1, 2, \dots, k.$$

(convince yourself)

$$\Rightarrow \langle y, u_i \rangle = \langle \hat{y}_S, u_i \rangle \quad i = 1, 2, \dots, k$$

$$\Leftrightarrow \begin{cases} \langle y, u_1 \rangle = \langle \hat{y}_s, u_1 \rangle \\ \langle y, u_2 \rangle = \langle \hat{y}_s, u_2 \rangle \\ \vdots \\ \langle y, u_k \rangle = \langle \hat{y}_s, u_k \rangle \end{cases} \quad \hat{y}_s \in S$$

Since $\hat{y}_s \in S$,

$$\hat{y}_s = \sum_{i=1}^k x_i u_i, \quad x_i \in \mathbb{F}$$

$$\langle y, u_j \rangle = \left\langle \underbrace{\sum_{i=1}^k x_i u_i}_{\hat{y}_s}, u_j \right\rangle$$

$j = 1, 2, \dots, k$

$$\langle y, u_j \rangle = \sum_{i=1}^k x_i \langle u_i, u_j \rangle$$

$j = 1, 2, \dots, k$

$$\langle y, u_1 \rangle = \overbrace{\left[\langle u_1, u_1 \rangle \langle u_2, u_1 \rangle \dots \langle u_k, u_1 \rangle \right]}^{1 \times k} x$$

where $x = \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix}$

Proceeding similarly, we obtain

$$\begin{bmatrix} \langle y, u_1 \rangle \\ \langle y, u_2 \rangle \\ \vdots \\ \langle y, u_k \rangle \end{bmatrix} = \begin{bmatrix} \langle u_1, u_1 \rangle & \langle u_2, u_1 \rangle & \dots & \langle u_k, u_1 \rangle \\ \langle u_1, u_2 \rangle & \langle u_2, u_2 \rangle & \dots & \langle u_k, u_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle u_1, u_k \rangle & \langle u_2, u_k \rangle & \dots & \langle u_k, u_k \rangle \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{bmatrix}$$

"Normal Equations"

If we can find $x \in \mathbb{R}^k$ satisfying
the Normal Equations, then
 $\hat{y}_s$ exists!

In particular $\hat{y}_s$ can be
constructed as

$$\hat{y}_s = \sum_{i=1}^K x_i u_i$$

• $\hat{y}_s$ exists $\Leftrightarrow$ Normal Equations
has a solution.

• $\hat{y}_s$ can be found by solving
the Normal Equations.

Do Normal Equations Always Have Solution?

$$\begin{bmatrix} \langle y, u_1 \rangle \\ \langle y, u_2 \rangle \\ \vdots \\ \langle y, u_k \rangle \end{bmatrix} = \begin{bmatrix} \langle u_1, u_1 \rangle & \langle u_2, u_1 \rangle & \dots & \langle u_k, u_1 \rangle \\ \vdots & \vdots & & \vdots \\ \langle u_1, u_k \rangle & \langle u_2, u_k \rangle & \dots & \langle u_k, u_k \rangle \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{bmatrix}$$

$k \times k$ matrix $\in \mathbb{F}^{k \times k}$ ($\mathbb{F} = \mathbb{C}$ or $\mathbb{R}$)
 $\triangleq \tilde{U}$

Hint:

Consider a vector $z \in \mathbb{F}^k$ (or $\mathbb{R}^k$)

When does

$$z^H \tilde{U} z = 0 \quad ?$$

[Use ^{general} properties of inner product $\langle \cdot, \cdot \rangle$]

You will be able to show that $\tilde{U}$ is full rank $\Rightarrow$ Normal Equations always have solution.

$\Rightarrow \hat{y}_s$ exists!

In particular, it can be found by first solving for $x_1, x_2, \dots, x_k$ in the Normal Equations.

And then construct K .

$$\hat{y}_s = \sum_{i=1}^K u_i x_i$$

How Does Orthogonal Projection Help Solve (P)?

Given $y \in \mathcal{V}$, $S \subseteq \mathcal{V}$,

$$\min_{v \in S} \|y - v\|_{ip}. \quad \text{--- (P)}$$

Theorem: The orthogonal projection of y onto S , denoted by $\hat{y}_S$, is the unique minimizer of (P).

Proof:

Consider any vector $v \in S$

$$\begin{aligned} \|y - v\|_{ip}^2 &= \|y - \hat{y}_S + \hat{y}_S - v\|_{ip}^2 = \langle y - \hat{y}_S + \hat{y}_S - v, y - \hat{y}_S + \hat{y}_S - v \rangle \\ &= \|y - \hat{y}_S\|_{ip}^2 + \|\hat{y}_S - v\|_{ip}^2 \\ &\quad + \underbrace{\langle y - \hat{y}_S, \hat{y}_S - v \rangle}_{\substack{\in S \\ (\text{since } \hat{y}_S \in S, \\ v \in S)}} + \underbrace{\langle \hat{y}_S - v, y - \hat{y}_S \rangle}_0 \end{aligned}$$

Since $\hat{y}_S$ is orthogonal projection of y onto S ,

$$\Rightarrow \langle \hat{y} - \hat{y}_S, \hat{y}_S - v \rangle = 0, \quad \forall v \in S.$$

$$\Rightarrow \|y - v\|_{ip}^2 = \|y - \hat{y}_S\|_{ip}^2 + \underbrace{\|\hat{y}_S - v\|_{ip}^2}_{\geq 0}$$

$$\Rightarrow \|y - v\|_{ip}^2 \geq \|y - \hat{y}_S\|_{ip}^2 \quad (\text{norm is non-negative})$$
$$\forall v \in S.$$

Equality holds if and only if

$$\|\hat{y}_S - v\|_{ip}^2 = 0$$

$$\Leftrightarrow \hat{y}_S - v = 0 \Leftrightarrow \hat{y}_S = v$$

$\hat{y}_S$ is the unique minimizer of (P)

SUMMARY

The optimization Problem (P)

$$\min_{v \in S} \|y - v\|_p$$

- (i) has a solution (or minimizer)
- (ii) the solution is unique and given by the orthogonal projection of y onto S .
- (iii) We find the solution by solving the Normal Equations (which can always be solved).

LEAST SQUARES

Given $A \in \mathbb{C}^{m \times n}$, $y \in \mathbb{C}^m$

Consider solving (for x):-

$$y = Ax$$

$$y \notin R(A)$$



No solution

$$y \in R(A) \\ \text{rank}(A) = n$$



Unique solution

$$y \in R(A) \\ \text{rank}(A) < n$$



infinite solutions.

LEAST SQUARES

$$\text{If } y \notin R(A)$$

$$\min_{x \in \mathbb{R}^n} \|y - Ax\|_2 \quad \text{--- (LS)}$$

Cast as closest point to subspace:-

$$\min \|y - v\|$$

s.t. $v \in R(A)$

$$(v = Ax \text{ for some } x) -$$

Exercise: Using what you have learned so far about orthogonal projections, show that the solution to LS is given by solution to the following system of equations:-

$$A^H y = (A^H A) x$$

It can be shown that this system is always solvable. even if $A^H A$ is not full rank. (HW problem) -

PROJECTION MATRIX

- Suppose $A^H A$ is invertible (ie $\text{rank}(A) = n$).

then x can be uniquely determined by

$$x^* = (A^H A)^{-1} A^H y$$

In this case, the orthogonal projection of y onto $R(A)$, is given by

$$\begin{aligned} \hat{y}_s &= A x^* \\ &= A (A^H A)^{-1} A^H y \\ &\quad \underbrace{\hspace{10em}}_{P_{R(A)}} \end{aligned}$$

$P_{R(A)}$ is called the orthogonal projection matrix onto $R(A)$.

Given any vector $z \in \mathbb{C}^m$,
the projection matrix acts on z
and returns the orthogonal
projection of z onto $\mathcal{R}(A)$

$$\text{i.e. } P_{\mathcal{R}(A)} z = \hat{z}_S$$

• Additionally if columns of A
are orthonormal, i.e.

$$a_i^H a_j = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

then

$$P_{\mathcal{R}(A)} = AA^H$$

$$\left(\because A^H A = I \right)$$